Low-Titer Membrane Affinity Chromatography

Kareem Fakhfakh*[1], Adam Powell[2], Martyn Hulley[2], Faisal Uddin[2], Jon Coffman[1]

[1]Downstream Bioprocess Technology & Engineering Biopharmaceutical Development, Gaithersburg, US; [2]Purification and Process Sciences, Biopharmaceutical Development, Cambridge UK

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Abstract

An explosion of next generation antibody-based modalities has driven new healthcare opportunities. For example, Bi/multi-specific targeted antibodies and antibody fragments have been applied in cancer and immunotherapies.[1]

A broadening portfolio of such new modalities has increased and hastened R&D demands for those materials which are often expressed in transient cultures with low titers <1g/L[2].

So, to accommodate increased process intensification, affinity membrane chromatography was explored for product capture and evaluated for process mass intensity (PMI) – a sustainability metric.

Introduction

- Highly porous membrane-based supports coupled with an affinity ligands serve as a promising alternative for product capture with high flow-rate. Lower residence times are thus achievable compared to packed beds enabling higher process intensity.[3]
- Higher wash volumes are typically needed compared to packed beds increasing buffer consumption and PMI, computed here as kg buffer consumed per kg purified mAb product.
- This work extends previous evaluation of convective media[4] adding context of performance and PMI for low-titer applications.

Methods

Lifetime Studies
Following dynamic binding capacity (DBC) studies, monoclonal antibody (mAb) was loaded to capacity following a manufacturer's recommended method.

Every 10 cycles, performance metrics were calculated as a function of cycle to estimate useable lifespan and ligand loss. Specifically, recovery (%; step yield), monomer purity (%; from HPLC-SEC), cumulative protein A ligand lost, and impurity clearance (*i.e.,* log reduction value, LRV for host cell protein, HCP; and DNA)

Loading DoE: Titer, Loading Residence Time
Purified protein was overloaded on a 1mL membrane and its DBC10% was measured. Productivity was computed assuming a recovery of 90%. Protein was loaded with a titer between 0.2 and 2.0 mg/mL and loading residence time between 20 and 120 seconds.

PMI Modelling
Python was used to compute PMI for a range of load and process scenarios. For each combination of load scenario (titer, mass load, maximum allowed processing time), equipment type (membrane, packed bed), and available volume, outputs were computed.

The minimum volume equipment that could complete the processing in the given time was selected for every titer and outputs were computed: 1) equipment volume, 2) capacity utilization, 3) number of cycles, and 4) PMI were output. Assumptions included a recovery of 90% for each method, equal capacity loaded per cycle.

Lifetime Study and Loading Screen DoE Method

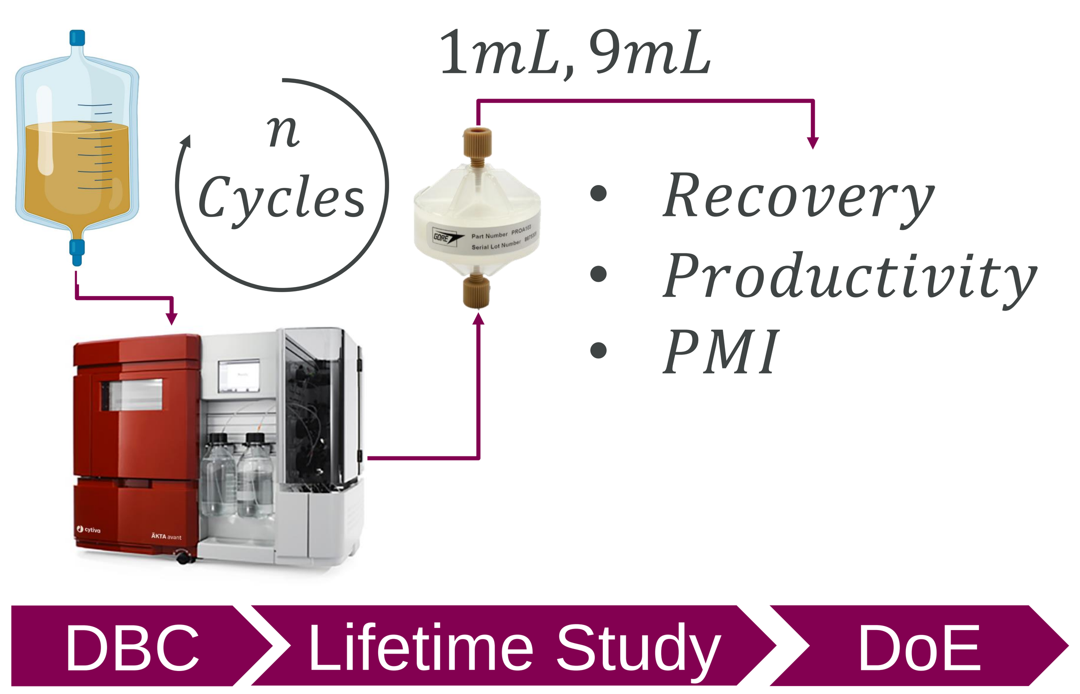

Figure 1. Method and lifetime study procedure overview. DBCs were first computed before a 100-cycle lifetime study, followed a screening of load conditions.

Results

Following manufacturer's recommendations for sanitization, we measured stable recovery and impurity clearance over 100 cycles in line with other reports.

When low-titer samples (0-2 mg/mL) are loaded between 20 and 120s residence time, productivity and DBC was computed. Though DBC10% increases ~30% with a slowed loading, productivity decreases by a higher magnitude from a longer loading time resulting in a trade off.

Takeaway: to maximize productivity for a given load titer, load at a low residence time.

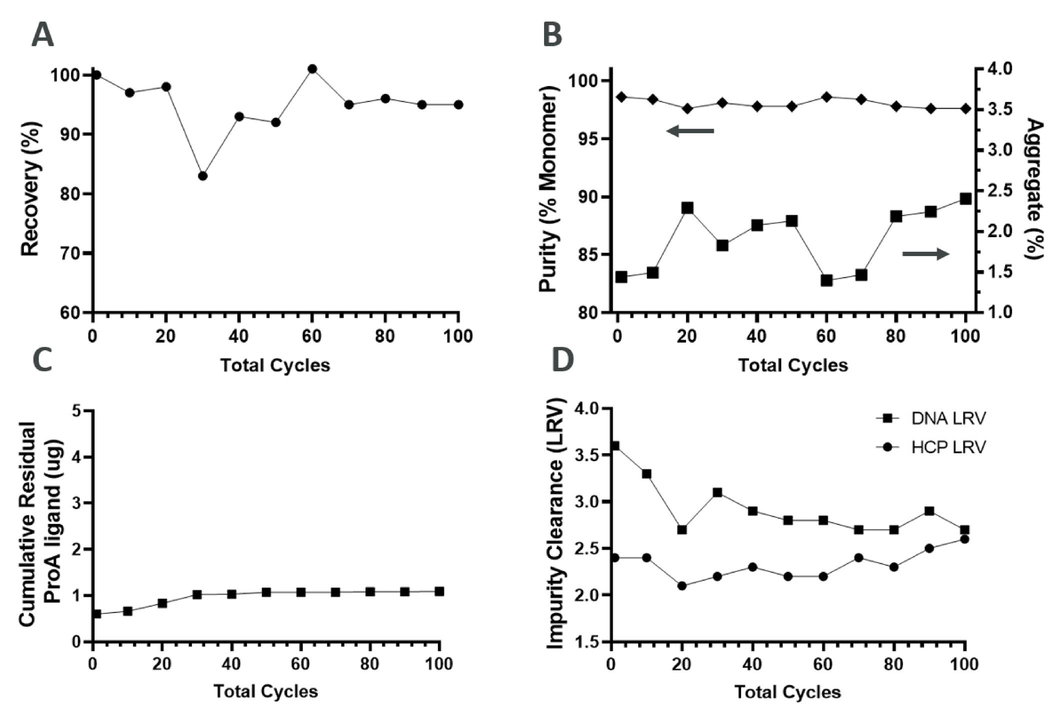

Figure 2. Recovery % (Panel A), HPSEC: TSKGel3000SwXL – derived monomer purity % and Aggregate content (Panel B; left and right respectively), cumulative residual protein A ligand (Panel C), Impurity clearance (Panel D; HCP and DNA LRV) on a 9 mL Gore® Protein A Capture device over 100 purification cycles.

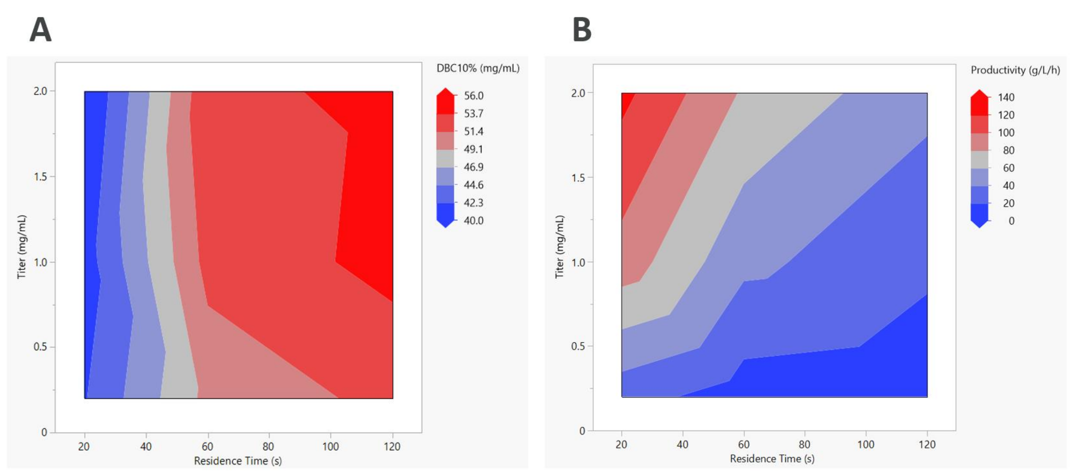

Figure 3. Contour plots of DBC10% (mg/mL; Panel A) and Productivity (g/L/h; Panel B) plotted against both loading residence time (s) and load titer (mg/mL).

Scenario Analysis Method

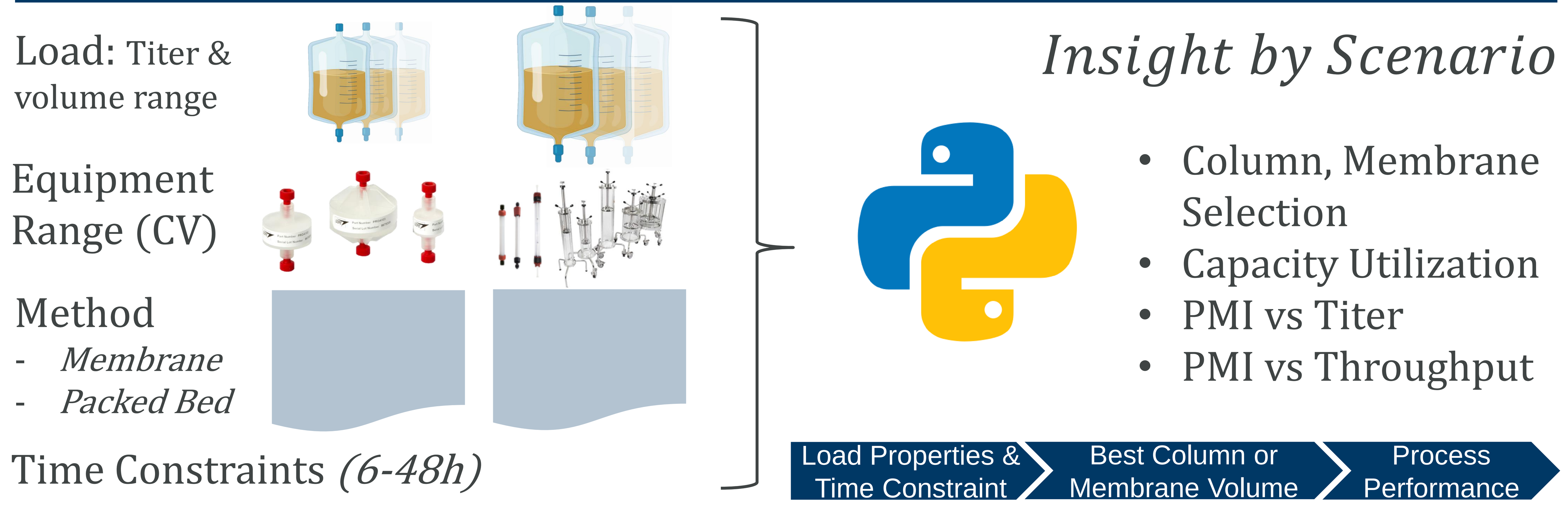

Figure 4. A range of process scenarios is explored having a range of load properties titer, mass (& volume), time constraints, discrete equipment volumes, and method settings. For each combination of inputs, column and membrane volumes are screened and the most efficient size for each scenario is evaluated.

Results

Given a load (*i.e.* volume, titer) and time constraint for purification, the *smallest* column/membrane volume required can be computed for a known purification method (*e.g.* flowrates, phase durations) and capacity for a given molecule.

Low-titers (*i.e.* high load time), exacerbate requirements for high flow-rate, thus large cross-sectional area, high-volume packed beds and few cycles. Without flexibility, low load mass can lead to low-capacity utilization (*i.e.* mass loaded / maximum possible load).

Membranes can accommodate higher flowrates better without decreasing capacity utilization since it has more flexibility to reduce its, generally higher, number of purification cycles required.

So, for packed beds below some titer (*ca.* <1g/L in this work), PMI suffers, increasing steeply when capacity utilization is low and process intensity is high. Above this titer, packed beds become more appropriately loaded and sized and use less buffer than membranes.

While membranes have higher productivity (g/L/h) compared to packed beds, they often use more cycles with a lower equipment volume. Plotting PMI vs throughput (g/h) accounts for these differences elucidating when a technology is preferred from a PMI perspective.

Takeaway:

- For a given purification time constraint and process assumptions, PMI depends on load titer where very low-titers can favor membrane adsorbers over packed bed columns despite higher wash volumes.

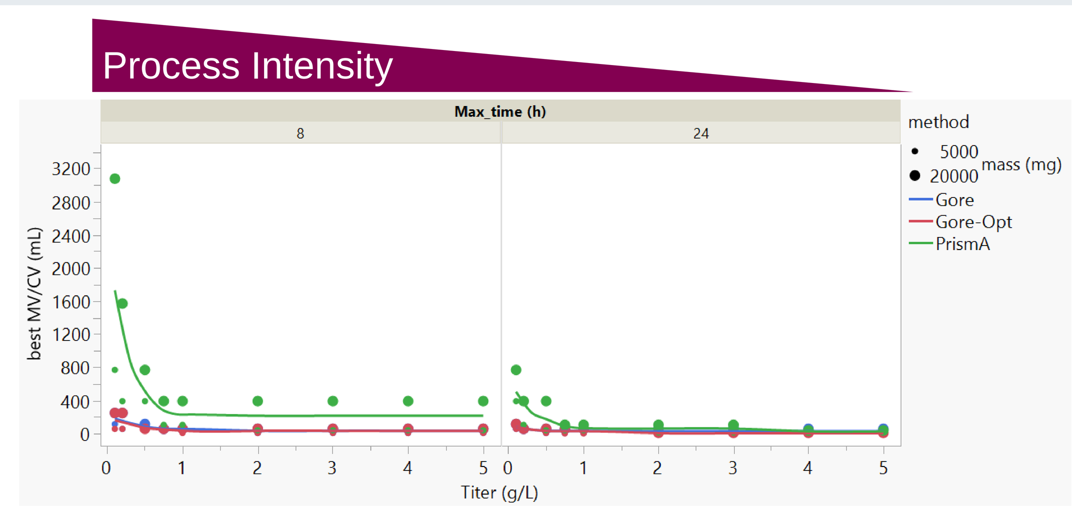

Figure 5. Minimum chromatography media volume plotted against titer for a set of maximum allowed processing times of 8 and 24 h (top axis) and a set of load masses equal to 5000 and 20,000 mg to be processed (dots). A spline passing through the mean y-value for all masses is shown as a line for each method to indicate the trend only. Membrane capacity is assumed to be 42.1*0.8 mg/mL. Methods include a Gore-recommended buffer optimized method (Gore-opt) a method with higher washes (Gore) and a packed bed high-capacity (80.0 mg/mL, assumed) method (PrismA).

Figure 6. Capacity utilization is plotted against number of cycles for each maximum processing time where colored markers denote the method type used for these calculations. As process speed decreases, capacity utilization decreases and generally, low numbers of cycles are concomitant with a decreasing capacity utilization.

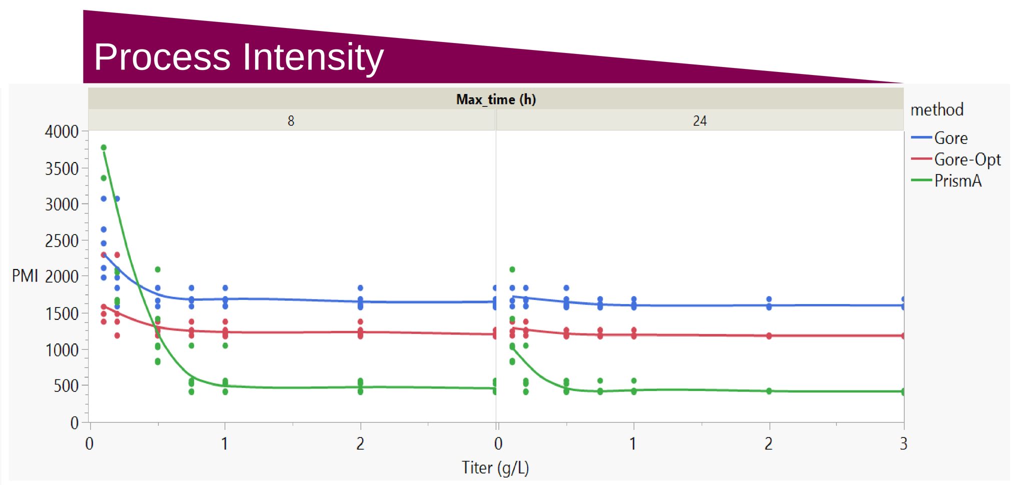

Figure 7. PMI plotted as a function of titer for a range of maximum processing times for 8 and 24 h, for each of the three methods. For every titer, a set of load masses was investigated. Pump washes were ignored, and membrane capacity was set to 42.1*0.8 mg/mL. Splines connect the mean PMI for each titer value colored by purification method.

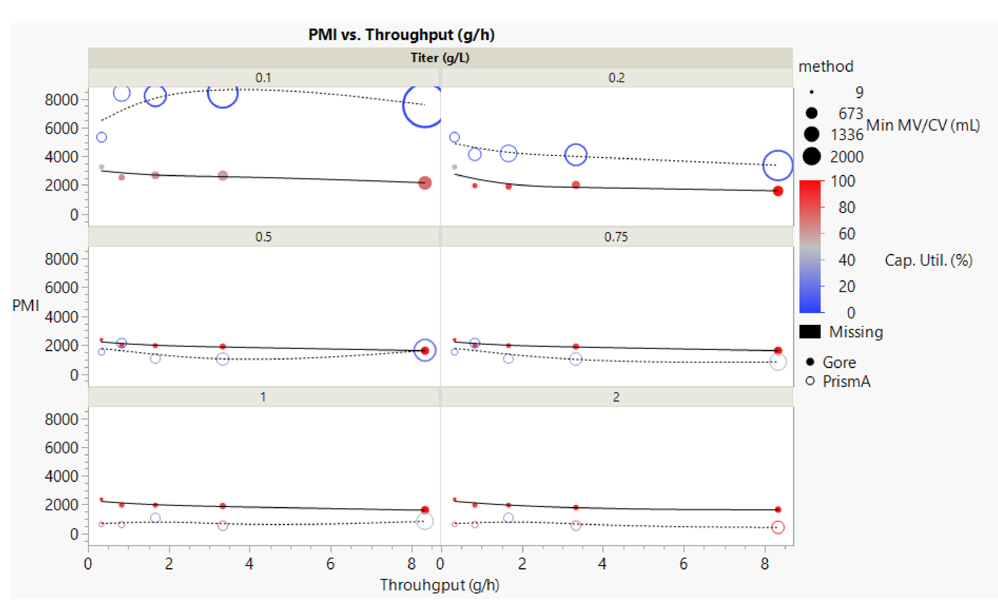

Figure 8. PMI plotted against throughput (g/h) for two affinity chromatography technologies: Gore® (filled circles, solid line) and MabSelect PrismA™ (open circles, dashed line). 42.1*0.8 mg/mL membrane capacity. Gore method is unoptimized, maximum processing time is 6h, pump wash volumes are counted.

Interactive Method Comparison WebApp

- To generalize the results to users' processes, an online tool was developed in Python Shiny.
- Users can tune two methods, load range, and processing speed to build and compare their scenarios.
- Features include:
  - Comparison of two custom methods.
  - Plotting of chromatography metrics as a function of titer.
  - Pump Wash contribution to PMIs.
- Outputs include:
  - PMI, Capacity utilization, best fit column volume, and number of cycles (not shown).

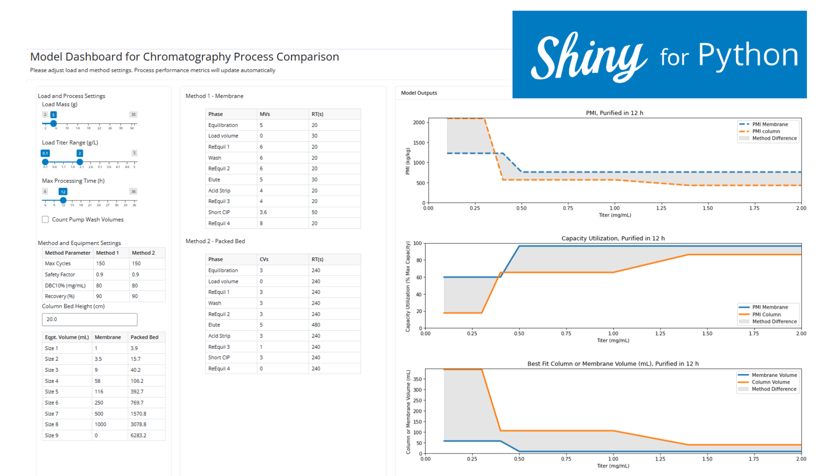

Figure 9. A web-based user interface shows a panel of controls for load and method inputs and a set of graphical outputs of sizing and performance. The graphical outputs describe interplay between the various sizing and performance attributes.

Conclusions

- Membrane performance (recovery, purity, impurity clearance) is stable over 100 cycles.
- Despite small capacity gains from slow loading, productivity is maximized in membrane chromatography at higher flowrates.
- High process intensity at low-titer tends to favor membrane use given their lower PMI for equal mass throughput (g/h). In our work, this effect disappears at higher titer (*ca.* >1g/L) and with more time allowed for purification (*ca.* >24h).
- Plotting PMI as a function of throughput and titer accounts for better comparison of processing scenarios with different equipment size and utilization.
- A web-based Python Shiny application was developed for chromatography users to compare their own in-house methods with membranes.

References

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